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Course: Game Programming

Question No: 01

Write any 2 advantages of new input system?

Advantages of the New Input System

The new Input System Package in Unity is a significant upgrade over the legacy system. Here are two primary advantages:

- **Action-Based Architecture & Easy Rebinding:** Unlike the old system which required you to "poll" for specific keys (e.g., `Input.GetKeyDown(KeyCode.Space)`), the new system uses **Actions**. This means you define an action like "Jump" and then map multiple devices to it. This makes it incredibly easy to implement **in-game control remapping** for players, as you are simply changing a binding to an existing action rather than rewriting code.
- **Event-Driven Workflow:** The new system is event-based rather than polling-based. Instead of checking for input every single frame in an `Update()` loop (which can be inefficient), your code "subscribes" to an event that only triggers when a button is actually pressed. This leads to much **cleaner, more organized code** and better performance, especially in complex projects.

Question No: 02

Describe usecase of overlay canvas,screen space canvas and world space canvas from any of your favorite game?

Canvas Use Cases (Example Game: *League of Legends*)

To illustrate how different Canvas types function, let's look at a popular game like **League of Legends**:

- **Overlay Canvas (Screen Space - Overlay):**
 - **Use Case:** This is used for the main **Heads-Up Display (HUD)**.
 - **Description:** The mini-map in the corner, your ability icons at the bottom, and your character's stats are all part of an Overlay Canvas. These elements are drawn directly onto the screen as a final layer. No matter where you move the camera or what is happening in the 3D world, these UI elements stay perfectly fixed and flat on your monitor.
- **Screen Space Canvas (Screen Space - Camera):**
 - **Use Case:** Used for **full-screen visual effects or "immersive" UI**.
 - **Description:** When your character takes a lot of damage and the edges of the screen flash red, or when the screen turns gray upon death, these are often handled by a Screen Space - Camera canvas. Because this canvas is "placed" a certain distance in front of a specific camera, it can interact with post-processing effects (like blur or color grading) more naturally than a standard overlay would.
- **World Space Canvas:**
 - **Use Case:** Used for **Health Bars and Player Names** above champions.
 - **Description:** When you see a health bar floating directly over an enemy's head, that is a World Space Canvas. Unlike the HUD, these UI elements exist "inside" the 3D world. As the champion moves around the map, the health bar moves with them. If the champion moves further away from your camera, the health bar gets smaller, respecting the 3D perspective of the game world.